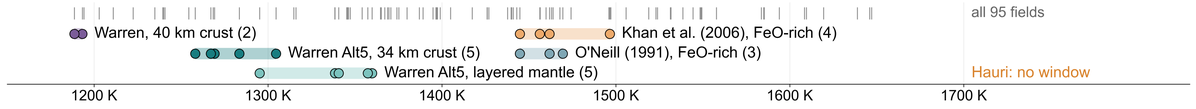
Thermochemical Models of the Lunar Mantle and Implications for Geophysical Observations

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No current model matches both temperature and composition to geophysical constraints.

Here is what we get: 19 interiors match mass, moment of inertia and core size, each composition only within its own window of mean mantle temperature



Takeaways

1 Mass, moment of inertia and core size cannot distinguish the compositions.
Each composition fits these data only within its own range of mantle temperature: **19 of 1,425** interiors match all three.

2 No tested composition reproduces the Apollo seismic models.
Composition shifts S-wave speed by **≤ 2%**; every interior is too slow at 500–900 km, 15 of 19 too fast at 60–270 km.

3 For tides, where the mantle is warm matters more than how warm.
Across the 19 interiors, k_{3m} varies by **2.5–2.8×**; **2 of 19** fit GRAIL's degree-1 structure; none of them fits the Apollo heat flow.

Approach

How. Forward models, not an inversion. Inputs: published bulk compositions (homogeneous, layered and FeO-rich mantles) and 95 published 3-D temperature fields (Santangelo et al. 2025). Every combination that matches the lunar mass gives an interior (crust extraction, mineralogy, radial structure): 1,425 in all. Outputs: moment of inertia, core size, seismic velocities and tidal Love numbers, compared with observations the interiors were not fitted to.

What is constrained. The Moon's mass, moment of inertia, gravity field and tidal Love numbers are measured; its bulk composition, temperatures, mineralogy and radial structure are inferred, not measured, and the Apollo seismic models rest on four nearside stations. This study explores that space; it does not determine a unique interior.

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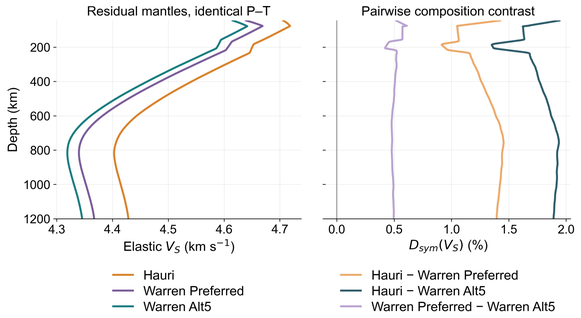
Bulk composition — Hauri et al. 2015 · Warren 2005
→ **Crust extraction** — mass balance → residual mantle
→ **Mineralogy** — Perple_X, Connolly 2009
→ **Radial structure** — PlanetProfile, Styczinski et al. 2023
→ **3-D temperature** — 95 fields, Santangelo et al. 2025
→ **Seismic constraints** — velocities vs Apollo models
→ **Tidal response** — LOV3D, Rovira-Navarro et al. 2024

A Reference states

1 Composition at identical P–T

Three residual mantles (Hauri et al. 2015; Warren 2005, two variants) at the same P and T (Stixrude & Lithgow-Bertelloni 2024).

Composition alone changes S-wave speed by ≤ 2%: Hauri is 0.9–1.9% faster than the two Warren mantles (0.4–0.6% apart).

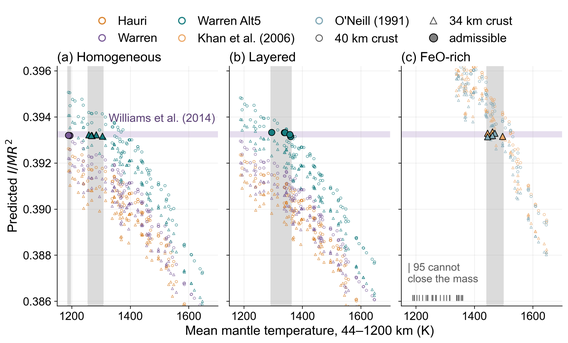


Not complete interiors. D_{sym} = 200 (A−B)/(A+B); 50-km averages across phase changes.

2 Which interiors fit?

Homogeneous, layered and FeO-rich mantles built to match the lunar mass (PlanetProfile; Styczinski et al. 2023) under 95 published 3-D temperature fields.

19 of 1,425 interiors match the moment of inertia and core radius, each mantle only in its own temperature window: cold for homogeneous mantles (7, a), warmer for layered (5, b) and FeO-rich ones (7, c).



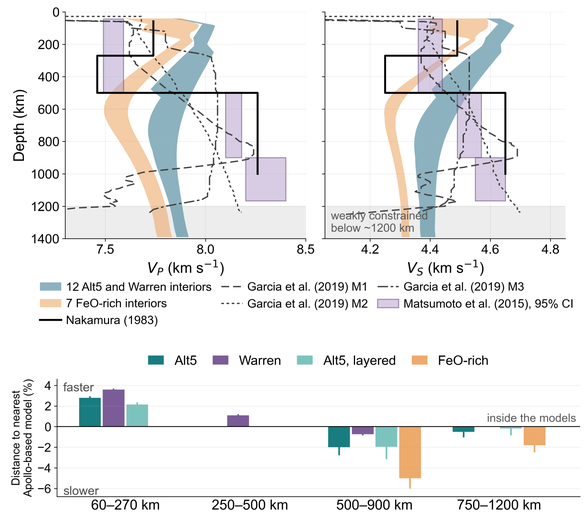
Moment of inertia: Williams et al. (2014); core 330 ± 20 km: Weber et al. (2011); the 19 have cores of 318–345 km. (b) 500-km magma ocean; (c) Khan et al. (2006), O'Neill (1991), Sossi et al. (2025); 95 cannot close the mass. 2 of the 19 match the Apollo heat flow (Langseth et al. 1976; Warren & Rasmussen 1987).

B Seismic implications

3 Comparison with Apollo-based models

The 19 interiors against Apollo-based models (Garcia et al. 2019; Matsumoto et al. 2015; Nakamura 1983).

Every interior is too slow at 500–900 km (0.7–5.9%); at 60–270 km, 15 are too fast (up to 3.6%) and four FeO-rich lie inside the models: FeO-rich mantles fix the top, worsen the bottom.

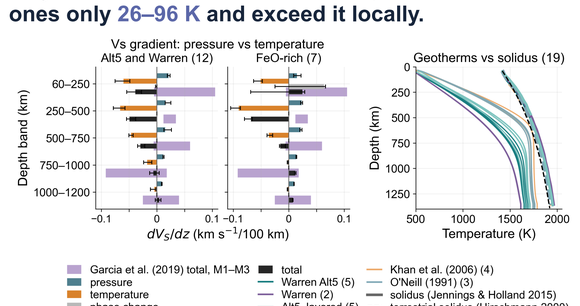


Profiles averaged over 50 km. Misfit = outside every Apollo-based model. Bar: closest interior of the group; whisker: farthest.

4 What sets the gradient, and how hot?

S-wave gradient split into pressure and temperature parts; dry solidus (Jennings & Holland 2015).

Down to 750 km, temperature acts 1.3–6× more than pressure in all 19. Homogeneous and layered mantles stay 170–350 K below their solidus; the hot FeO-rich ones only 26–96 K and exceed it locally.

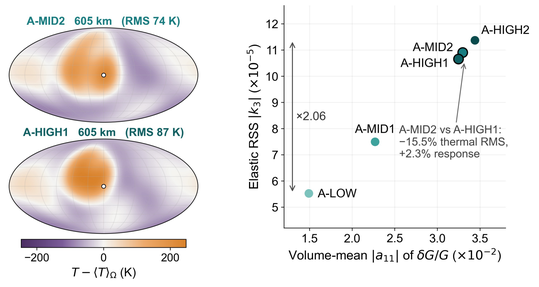


FeO-rich: steps across a phase change shown apart; solidi are upper bounds (no Na_2O in their compositions). Lunar residual mantles melt 31–45 K hotter than terrestrial peridotite (Hirschmann 2000).

C Tidal implications and outlook

5 Lateral geometry, not only amplitude

Five Alt5 interiors, same radial structure: the degree-3 response varies 2.06×, set by the degree-1 pattern, not the thermal amplitude.

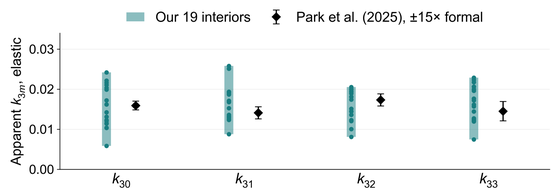


Maps at 605 km depth. RSS $|k_3|$: response to a unit forcing.

6 What would GRAIL measure?

What GRAIL would measure for each interior: 2012 orbit (Park et al. 2021), fit of Park et al. (2025).

Park et al.'s k_{3m} fall within the range of the 19 interiors, in all four orders.



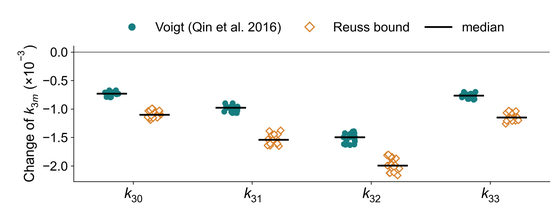
Lowest k_{30}: one FeO-rich interior with an inverted degree-1 pattern (μ_{11} = −1.1%).

Our code reproduces Park et al.'s own forward test (their Fig. 1a) to 0.94–0.99×.

7 Our model: the GRAIL crust

GRAIL crustal thickness (Wieczorek et al. 2013) mapped to shear modulus (Qin et al. 2016), 19 interiors.

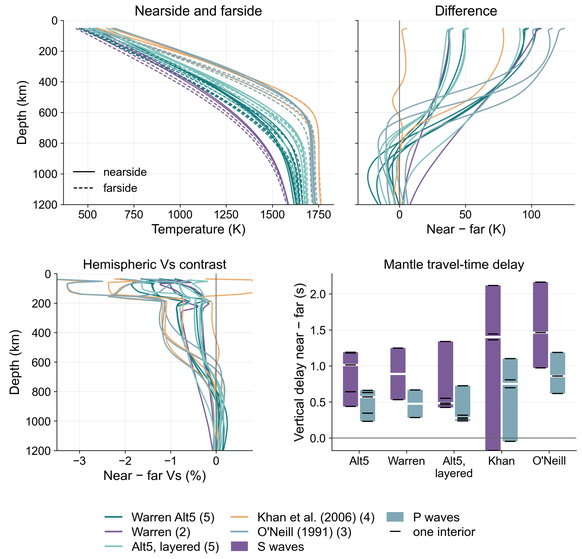
The crust lowers k_{3m} by only 0.0007–0.0016, about one Park et al. error bar: the mantle sets k_{3m}.



Change from a uniform crust. Voigt: equal strain; Reuss: equal stress.

8 Nearside versus farside

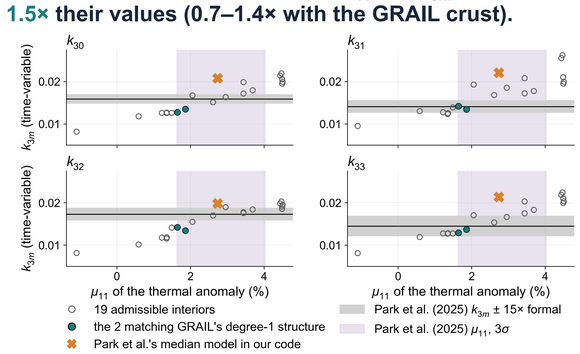
In 18 of 19, the nearside is 39–125 K warmer and 0.5–3.3% slower at 60–160 km; vertical S-wave delays range from −0.2 to 2.2 s between equally good fits.



Near-minus-far averages of the 3-D fields (mantle only), not ray paths; uniform crust. Bars: range of a family, white: median.

9 GRAIL selects the geometry

k_{3m} follow μ_{11}, the degree-1 amplitude: only 2 of 19 interiors fit GRAIL's degree-1 structure, none fits the Apollo heat flow; at Park et al.'s μ_{11} our k_{3m} are 0.8–1.5× their values (0.7–1.4× with the GRAIL crust).



19 admissible interiors: 7 homogeneous, 5 layered, 7 FeO-rich (panel 2). Degree-1: μ_{11} and μ_{10} within the 3σ of Park et al. (2025). ×: their median model in our code.

→ Composition changes S-wave speed by **≤ 2%** but sets which temperatures fit: **radial data cannot determine the composition.**

→ No composition brings the interiors inside the Apollo-based models: temperature sets the gradient.

→ Tides follow where the mantle is warm: across the 19, k_{3m} varies by 2.5–2.8×.
→ 2 of 19 fit GRAIL's degree-1 structure; none fits the Apollo heat flow.

Assumptions: dry solid equilibrium (locally exceeded in the hot FeO-rich mantles); residual mantle uniform or layered at 500 km; prescribed crust; liquid Fe–5 wt% S core; imposed 3-D temperature fields common to all compositions; radial pressure; no lateral density in the tidal solver (first-order estimate for the crust: ≤ 0.001 on k_{3m}).

References: Connolly (2009) G³; Garcia et al. (2019) SSR; Hauri et al. (2015) EPSL; Hirschmann (2000) G³; Jennings & Holland (2015) J. Petrol.; Khan et al. (2006) JGR; Langseth et al. (1976) Proc. LSC 7; Matsumoto et al. (2015) GRL; Nakamura (1983) JGR; O'Neill (1991) GCA; Park et al. (2021) AJ; Park et al. (2025) Nature; Qin et al. (2016) GJI; Rovira-Navarro et al. (2024) PSJ; Santangelo et al. (2025) JGR Planets; Sossi et al. (2025) Treatise Geochem.; Stixrude & Lithgow-Bertelloni (2024) GJI; Styczinski et al. (2023) ESS; Warren (2005) MAPS; Warren & Rasmussen (1987) JGR; Weber et al. (2011) Science; Wieczorek et al. (2013) Science; Williams et al. (2014) JGR Planets.